Ultra-Scale Neuromorphic-based Optimisation

A new generation of low latency, low power, and small hardware footprint optimisation algorithms

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The energy problem I

NeurOpt

Modern AI is consuming energy at a rate we cannot sustain!!!

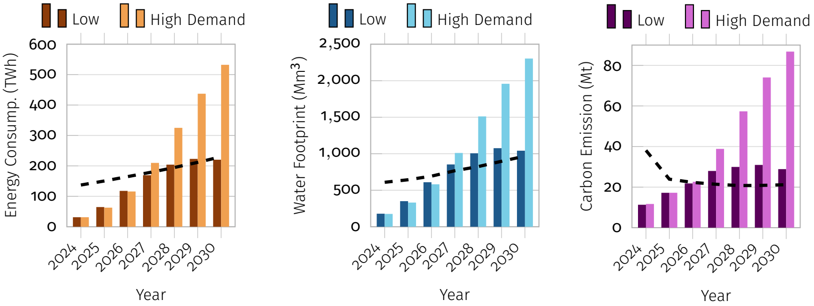


Figure: The energy consumption (left), water footprint (centre), and carbon emissions (right) of AI servers from 2024 to 2030 under two deployment scenarios from Low to High Demand.

The dashed stroke (reference) indicates the projected footprint of the entire US data-centre sector from prior literature.

Dataset was taken from Xiao et al. (2025).

( ) Energy Consump. ( ) Water Footprint ( ) Carbon Emission

T. Xiao et al., "Environmental impact and net-zero pathways for sustainable artificial intelligence servers in the usa", Nature Sustainability, 1–13 (2025)

The energy problem II

NeurOpt

- Training a single **large AI model** now emits as much CO_2 as five transatlantic flights.
- GPUs run at **constant full power** regardless of their activity, wasting energy at idle.
- Solving large optimisation problems with today's tools is costly, slow & energy-hungry.
- IoT and embedded systems require on-device inference with milliwatt budgets; impossible with today's GPU/CPU-centric optimisation.

The gap in existing work

Neuromorphic chips exist, but the algorithms are still classical.

- Most current neuromorphic systems use brain-inspired hardware but run conventional, gradient-based software on top of it[1].
- Evolutionary and swarm intelligence algorithms (proven solvers for hard optimisation) have never been designed natively for spiking hardware[2].
- No framework yet unifies algorithm design, scalability, and hardware co-optimisation into a single coherent approach.

[1] Sanaullah et al., "Exploring spiking neural networks: a comprehensive analysis of mathematical models and applications", Frontiers in Computational Neuroscience **17**, 1215824 (2023).

[2] T. Sasaki and H. Nakano, "Swarm intelligence algorithm based on spiking neural-oscillator networks, coupling interactions and search performances", Nonlinear Theory and Its Applications, IEICE **14**, 267–291 (2023).

What is NeurOpt?

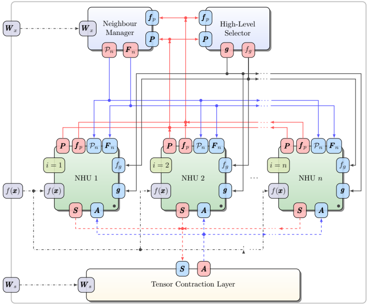
Optimisation that emerges from brain-like spiking activity.

- NeurOptimisers are algorithms where the search for an optimal solution is carried out by a network of spiking neurons, not a conventional loop.
- Each neuron fires a spike that triggers a small search move; collectively, many neurons explore and converge on a solution.
- The result is fully event-driven: computation only happens when there is something to compute, drastically reducing energy use.

What we have been doing so far

NeurOptimiser

Neuromorphic Optimiser



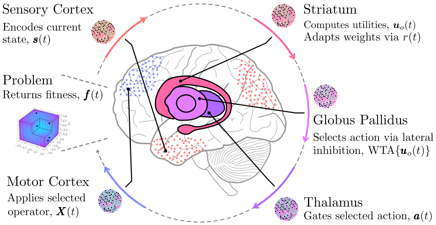
Paradigm: Comm. Sequential Processes (CSP)

Framework/Target: Lava NC/Intel Loihi 2

Sub-watt perf. (≤ 1.35 W), Linear runtime scaling

(NEVO)

Neuromorphic EVolutionary Optimiser



Paradigm: Neural Engineering Framework (NEF)

Framework/Target: Nengo/Agnostic

First Spike-driven operator selection via CBTL: Proof-of-concept validated

NeurOptim: Our research initiative

- NeurOptim[a] is a research initiative exploring how **Neuromorphic Computing** and **Computational Intelligence** can give rise to a new class of **Optimisation Algorithms**.
- Collective progress:
  - Open Science
  - Transparent methodologies
  - Reproducible experimentation

[a] `https://neuroptim.org`

Publications

NeurOpt

6 publications: 2 accepted, 2 published, 2 preprint/under review.

Publications

- Cruz & Talbi, Towards Neuromorphic Evolutionary Computation: A Position Paper, IEEE WCCI/CEC [Accepted], 2026.
- Cruz & Talbi, NEVO: A Neuromorphic EVolutionary Optimiser with Spike-Driven Cortico-Basal-Thalamic Coordination, IEEE WCCI/CEC [Accepted & **Nom. to Best Paper**], 2026.
- Cruz & Talbi, NeurOptimiser: A General Framework for Neuromorphic Optimisation, WBO [Published], 2025.
- Talbi & Bouvier, NEVA: A Neuromorphic Evolutionary Algorithm, OLA [Published], 2025.
- Cruz & Talbi, NeurOptimisation: The Spiking Way to Evolve, ArXiv/IEEE TEVC [UR], 2025.
- Talbi, Neuromorphic-based MHs: A new generation of low power, low latency and small footprint optimization algorithms, ArXiv/Elsevier Journal [UR], 2025.

Events, tools & more

NeurOpt

An open ecosystem: tutorials, sessions, datasets, and frameworks.

Events

- Special Session on Neuromorphic Computing, OLA, 2026.
- Special Session on Neuromorphic and Computational Intelligence (NeuCompI): Synergies, Applications, and Learnings, IEEE WCCI/CEC, 2026.
- Tutorial: Neuromorphic Evolutionary Computation (NEC), IEEE WCCI/CEC, 2026.

Tools

- NeurOptimiser: A neuromorphic optimisation framework in which metaheuristic search emerges from asynchronous spiking dynamics. `https://neuroptimiser.github.io`
- NEVO: Neuromorphic Evolutionary Optimisation. A Python framework that bridges evolutionary computation and neuromorphic computing. `[Coming soon]`

What are we doing right now? (Overview)

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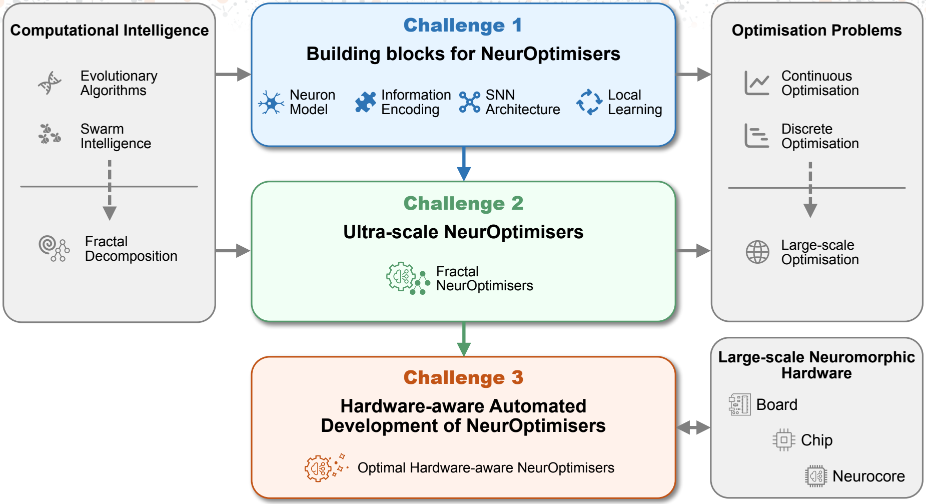

Phase 1: Building the foundations

NeurOpt

Can we design versatile, brain-inspired optimisers?

Objective

Design and simulate the **core building blocks** of NeurOptimisers.

Design questions

- Which neuron model best balances accuracy and energy cost?
- How do we encode an optimisation problem into spike patterns?
- What network shape best supports exploration and exploitation?
- Can neurons learn locally, without central supervision?

Success threshold

Consistent performance across standard BBOB problems, at least 75% functional coverage

Phase 2: Scale up

NeurOpt

Can NeurOptimisers handle problems with hundreds of variables?

Objective

Develop Ultra-scale NeurOptimisers using fractal and co-evolutionary mechanisms.

Two new ideas

- Fractal NeurOptimisers: hierarchical, self-similar network structures that recursively divide large problems into manageable sub-problems (as the brain sorts complexity).
- Co-evolutionary NeurOptimisers: multiple interacting populations of SNNs that collaborate and compete, collectively solving problems too large for any single network.

Success threshold

Competitive performance on problems with 160D to 640D, with search time growing slower than quadratically with problem size.

Phase 3: Connect to hardware

NeurOpt

Can an algorithm and a chip be designed together as one system?

Objective

Hardware-aware automated development.

Three sub-problems

- Find the best network shape: auto-search over neuron types, connectivity, etc.
- Find the best hardware config.: map the spiking network onto chips like Intel Loihi 2 (3?) or SpiNNaker2 to minimise energy and communication.
- Optimise both together: use MOO Algs. to find the best algorithm–hardware trade-off.

Target performance

ms-level iteration speed & at least 20× lower energy than equivalent CPU/GPU setups.

Expected impact

NeurOpt

A new field, new tools, and a new energy paradigm for optimisation.

Scientific Impact	Community Impact	Practical Impact
First principled framework connecting evolutionary computation and neuromorphic hardware	Open tools, datasets, tutorials, and summer schools	20× or more energy reduction over GPU-based optimisation
Publications at top venues (WCCI, GECCO, NeurIPS)	Special sessions at OLA, WCCI, etc., building the research community	Scalable to hundreds of variables on real neuromorphic chips
		Deployable on edge devices (IoT sensors, embedded controllers) where cloud offloading is infeasible.

NeurOpt builds the algorithmic, computational, and hardware foundations for a new generation of low-power, scalable, brain-inspired optimisation.

Thanks!
Grazie!
Merci!
Mulţmesc!
¡Gracias!

NeurOptim

neuroptim.org

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